PHARMACOVIGILANCE INTERNSHIP

TASK 4

CASE CLASSIFICATION & SEVERITY ANALYSIS
Seriousness and Expectedness Assessment

Prepared by: Yogesh Kumar
Role: Pharmacovigilance Intern
Programme: B.Pharm | Raj Kumar Goel Institute of Technology (AKTU)
Case status: Fictional educational training cases

Project objective

To classify adverse drug reactions along two independent dimensions - seriousness and expectedness - using a structured pharmacovigilance approach. The assessment focuses on ICH E2A seriousness criteria, reference safety information, clinical evidence, documentation quality and the regulatory implications of the combined classification.

> **Continuity with Tasks 1-3:** the project carries forward the earlier ADR-identification and documentation skills and applies them to a new classification layer. The ibuprofen hypersensitivity scenario used in Tasks 1-2 is retained as the continuity case here, while the remaining cases are separate fictional training scenarios.

Important declaration

> All patients, dates, reporters and case details in this educational portfolio are fictional. This document is not a real case report, medical advice or a substitute for an approved pharmacovigilance procedure. Expectedness is a labeling-based determination and must be checked against the current, version-controlled reference safety information in real-world work.

1. Learning objectives

By the end of this exercise, the assessor should be able to: (1) distinguish seriousness from symptom severity; (2) apply the ICH E2A seriousness criteria; (3) determine expectedness against reference safety information; (4) document the evidence for each criterion; (5) identify cases that are serious and unexpected; and (6) explain why a case may be serious but expected or non-serious but unexpected.

2. ICH E2A seriousness criteria

Criterion	Operational meaning in the case	What must be shown
Death	The event results in death.	Documented fatal outcome.
Life-threatening	The patient was at immediate risk of death as the event occurred.	Clinical assessment of immediate threat.
Hospitalization	The event requires inpatient admission or prolongs an existing admission.	Admission/prolongation attributable to the event.
Disability / Incapacity	Persistent or significant disability or incapacity results.	Functional impact that meets the seriousness threshold.
Congenital anomaly	Congenital anomaly or birth defect in offspring.	Applicable pregnancy/exposure context and outcome.
Other medically important	Does not meet the above but may jeopardize the patient or require intervention to prevent a listed outcome.	Appropriate medical judgement supported by evidence.

Key rule

Seriousness is an outcome-based regulatory concept. A symptom can be severe or distressing yet remain non-serious if none of the seriousness criteria are met. Conversely, a clinically familiar event can still be serious when it results in hospitalization, life-threatening illness, disability or another qualifying outcome.

3. Expectedness determination

Expectedness asks whether the nature, severity and specificity of the reaction are consistent with the applicable reference safety information (RSI), such as an approved label, package insert or company core safety document. It is therefore a document-driven assessment rather than a judgement of how common, plausible or clinically familiar the reaction feels.

4. Evidence checklist for classification

Question	Seriousness relevance	Expectedness relevance
Did the event result in death?	Direct seriousness criterion.	Does not itself determine expectedness.
Was there an immediate threat to life?	Direct seriousness criterion.	Does not itself determine expectedness.
Was hospitalization required/prolonged?	Direct seriousness criterion.	Does not itself determine expectedness.
Was there persistent/significant disability?	Direct seriousness criterion.	Does not itself determine expectedness.
Was a congenital anomaly involved?	Direct seriousness criterion.	Check whether the nature/specification matches RSI.
Was another medically important intervention needed?	May qualify under other medically important.	Still requires separate RSI comparison.
Is the reaction listed in RSI?	Not a seriousness criterion.	Primary expectedness question.
Does the observed severity/specificity match RSI?	Not a seriousness criterion.	A mismatch can make a listed reaction unexpected.

5. Regulatory logic

For the training framework used in this portfolio, the combined classification is interpreted as follows: non-serious/expected and non-serious/unexpected cases remain differentiated but do not meet the combined serious-and-unexpected condition; serious/expected cases remain serious but are not treated as expedited solely because they are serious; serious/unexpected cases are the highest-priority combination for expedited processing.

> **Important:** This report demonstrates the classification logic from the supplied Task 4 training material. Real reporting timelines depend on the applicable jurisdiction, reporting rules, product, authority and current guidance.

6. Connection to earlier work

Task 1 established ADR identification, AE-versus-ADR distinction, reaction pattern classification and causal reasoning. Task 2 structured those observations into a reporting-style record. Task 3 then evaluated causal strength using WHO-UMC-style reasoning. Task 4 adds a separate regulatory classification layer: *how serious is the outcome?* and *is the reaction expected from the RSI?* Those questions are answered independently.

CASE 1 - IBUPROFEN

Continuity case from Yogesh Kumar Tasks 1-3

Patient: 35-year-old woman with intermittent migraine and seasonal allergic symptoms.
Concomitant medicine: Cetirizine 10 mg once daily.
Suspected medicine: Ibuprofen 400 mg three times daily after food for bacterial pharyngitis.
Reaction: Pruritic, raised rash over both forearms and upper chest with mild swelling around the lips.

Case narrative

Ibuprofen was newly initiated. On Day 2 the patient developed the rash and mild lip swelling. No new food, cosmetic product, insect exposure or other newly started medicine was reported. Ibuprofen was discontinued. Symptoms began improving within approximately 24 hours and later settled. Rechallenge was not performed. This is the same fictional clinical thread documented in the earlier reporting exercise.

Seriousness assessment

ICH E2A criterion	Met?	Case evidence
Death	No	Recovery documented; no fatal outcome.
Life-threatening	No	No airway, respiratory or circulatory compromise was documented in the source case.
Hospitalization	No	No inpatient admission reported.
Disability / incapacity	No	No persistent functional impairment reported.
Congenital anomaly	No	Not applicable to the fictional case.
Other medically important	No	Symptoms improved with withdrawal; no intervention to prevent a serious outcome was reported.

Seriousness classification: NON-SERIOUS

Expectedness assessment

For this educational case, the rash/pruritus is treated as an **expected** hypersensitivity-type reaction because the supplied Task 4 framework treats familiar labeled reactions as expected. In real-world work, the exact current amoxicillin/ibuprofen product RSI would have to be checked; expectedness cannot be inferred from class effect alone.

Combined classification: NON-SERIOUS / EXPECTED

Learning point: a clinically notable skin reaction does not become serious simply because it is an ADR. The outcome criteria must be assessed independently. This classification also preserves the distinction between causal assessment (Task 3) and regulatory seriousness/expectedness assessment (Task 4).

CASE 2 - AMOXICILLIN

Suspect medicine: Amoxicillin 500 mg orally
Patient: 29-year-old adult treated for bacterial sinusitis
Reaction: Mild generalized pruritic rash without systemic involvement

Narrative

Two days after starting amoxicillin, the patient develops an itchy rash across the trunk and upper limbs. There is no swelling, breathing difficulty or circulatory compromise. The medicine is continued under observation. The rash resolves with symptomatic antihistamine treatment and no inpatient care is needed.

Seriousness assessment

Criterion	Met?	Rationale
Death	No	Patient recovers.
Life-threatening	No	No airway, breathing or circulation compromise.
Hospitalization	No	Outpatient management only.
Disability / incapacity	No	No lasting impairment.
Congenital anomaly	No	Not applicable.
Other medically important	No	Symptomatic management only; no intervention to prevent a qualifying serious outcome.

Seriousness classification: NON-SERIOUS

Expectedness

The training assumption is that mild rash and pruritus are described within the applicable reference safety information for amoxicillin. Therefore the case is classified as expected for this exercise.

Combined classification: NON-SERIOUS / EXPECTED

Pharmacovigilance significance: although the case does not trigger a serious classification, documenting the reaction contributes to routine safety monitoring and helps characterize reaction frequency and patterns.

Expectedness caution

In a real case, the assessor should compare the observed reaction's nature, severity and specificity against the version-controlled product RSI. A general statement that a class can cause rash is not, by itself, sufficient for product-specific expectedness.

CASE 3 - METFORMIN

Suspect medicine: Metformin 1000 mg orally twice daily
Patient: 68-year-old woman with type 2 diabetes and declining renal function
Reaction: Lactic acidosis with confusion, rapid breathing and abdominal pain

Narrative

The patient presents to an emergency department with confusion, rapid breathing and abdominal pain. Laboratory testing demonstrates markedly elevated lactate with metabolic acidosis. She is admitted to intensive care, metformin is discontinued and supportive treatment is provided. The treating team considers her to have been at immediate risk of death before stabilization. She recovers following prolonged inpatient management.

Seriousness assessment

Criterion	Met?	Rationale
Death	No	Patient survives.
Life-threatening	Yes	Immediate risk of death at presentation was documented by the treating team.
Hospitalization	Yes	ICU admission and prolonged inpatient care required.
Disability / incapacity	Not assessed	Not required once a seriousness criterion is clearly met.
Congenital anomaly	No	Not applicable.
Other medically important	Not assessed	Life-threatening and hospitalization criteria already establish seriousness.

Seriousness classification: SERIOUS

Expectedness assessment

For this training case, the event is treated as **unexpected** because the case specifies that its observed severity/population-specific presentation is not clearly captured by the reference safety information as assessed here. In real practice the exact current product label must be checked.

Combined classification: SERIOUS / UNEXPECTED

Regulatory significance: this is the key combination for expedited handling in the supplied training framework. The reason is not merely that the event is clinically severe; it is the combination of a qualifying serious outcome and a determination that the event is unexpected against RSI.

Risk-factor context

Declining renal function is an important patient factor in the narrative. It does not by itself determine expectedness, but it should be preserved in the case record because it may materially affect the clinical interpretation of the event.

CASE 4 - ATORVASTATIN

Suspect medicine: Atorvastatin 20 mg orally once daily
Patient: 52-year-old man treated for hyperlipidemia
Reaction: Mild asymptomatic elevation of liver transaminases

Narrative

Routine liver-function testing six weeks after atorvastatin initiation identifies a mild increase in transaminases, below three times the upper limit of normal. The patient has no jaundice, abdominal pain or systemic symptoms. Atorvastatin is continued at the same dose with repeat monitoring planned.

Seriousness assessment

Criterion	Met?	Rationale
Death	No	No fatal outcome.
Life-threatening	No	Asymptomatic laboratory finding; no immediate threat documented.
Hospitalization	No	Outpatient monitoring only.
Disability / incapacity	No	No functional impact.
Congenital anomaly	No	Not applicable.
Other medically important	No	No intervention to prevent a serious outcome was required in the scenario.

Seriousness classification: NON-SERIOUS

Expectedness assessment

For the training exercise, mild asymptomatic transaminase elevation is treated as expected because the case is framed as a recognized, labeled laboratory monitoring finding. The key assessment is the match between the observed specificity and the product RSI.

Combined classification: NON-SERIOUS / EXPECTED

Learning point: laboratory abnormalities can be safety-relevant without meeting regulatory seriousness criteria. The absence of symptoms or hospitalization does not make documentation unnecessary; it means the case is appropriately handled within its classification.

CASE 5 - LEVOFLOXACIN

Suspect medicine: Levofloxacin 500 mg orally once daily
Patient: 57-year-old man, recreational runner, treated for urinary tract infection
Reaction: Partial Achilles tendon rupture with temporary functional limitation

Narrative

Approximately ten days into treatment, the patient develops sudden Achilles tendon pain followed by difficulty bearing weight. Clinical examination and imaging confirm a partial tendon rupture. Levofloxacin is discontinued, orthopedic assessment is arranged and the patient receives a supportive boot and structured rehabilitation. No inpatient admission is required, but the injury results in a significant temporary limitation of normal function.

Seriousness assessment

Criterion	Met?	Rationale
Death	No	Patient survives.
Life-threatening	No	No immediate threat to life documented.
Hospitalization	No	Managed through outpatient orthopedics.
Disability / incapacity	Yes	Significant temporary functional limitation requiring rehabilitation.
Congenital anomaly	No	Not applicable.
Other medically important	Not assessed	Disability criterion already establishes seriousness.

Seriousness classification: SERIOUS

Expectedness assessment

For this training exercise, tendon rupture is treated as expected because the case describes it as a recognized fluoroquinolone risk documented in reference safety information. The exact product RSI should still be checked in real practice.

Combined classification: SERIOUS / EXPECTED

Learning point: a case can be serious because of disability even when it does not require hospitalization. Seriousness and expectedness remain separate axes.

5. Comparative classification of the five cases

Case	Drug	Reaction	Seriousness	Expectedness	Combined classificatior
1	Ibuprofen	Pruritic rash + mild lip swelling	Non-serious	Expected*	Non-serious / Expected
2	Amoxicillin	Mild pruritic rash	Non-serious	Expected*	Non-serious / Expected
3	Metformin	Lactic acidosis	Serious	Unexpected*	Serious / Unexpected
4	Atorvastatin	Mild asymptomatic transaminase elevation	Non-serious	Expected*	Non-serious / Expected
5	Levofloxacin	Partial Achilles tendon rupture	Serious	Expected*	Serious / Expected

*Expectedness labels are training-case determinations to be verified against the applicable, current product reference safety information in real pharmacovigilance work.

6. Comparative reasoning

Case 1 demonstrates a non-serious outcome carried forward from the earlier ADR-identification and reporting tasks. Case 2 demonstrates that a familiar cutaneous reaction may remain non-serious and expected. Case 3 shows why a case with a serious clinical outcome must then be checked independently for expectedness; when the event is also unexpected, it becomes the highest-priority combination in the supplied framework. Cases 4 and 5 reinforce that seriousness is not synonymous with hospitalization: a laboratory abnormality can remain non-serious, while a meaningful disability can qualify a case as serious.

7. Why seriousness and expectedness must remain separate

The assessor should not use one axis as a shortcut for the other. A serious event may be expected, as shown by the hospitalization or disability cases. Conversely, a mild reaction can be unexpected if its nature, severity or specificity is not represented in RSI. The final combined classification therefore comes only after both dimensions have been independently documented.

8. Documentation quality and submission checklist

Quality check	Status	Evidence in this portfolio
Patient/event scenario clearly described	Complete	Each case has a concise profile and narrative.
Seriousness criteria assessed separately	Complete	ICH E2A-style criterion tables provided for each case.
Expectedness assessed separately	Complete	Each case states the RSI-based training assumption.
Combined classification shown	Complete	Final label shown after independent assessment.
Continuity with Tasks 1-3 preserved	Complete	Ibuprofen case carries forward from earlier Yogesh submissions.
Missing data not silently invented	Complete	Real-world limitations and RSI verification caveats are stated.
Educational status visible	Complete	Fictional training declaration included.
YouTube/external tutorial links excluded	Complete	Submission document contains no tutorial link.

9. Practical pharmacovigilance takeaways

1. Start with the documented clinical outcome, not the perceived importance of the medicine.
2. Apply each ICH E2A seriousness criterion explicitly.
3. Do not equate severe symptoms with regulatory seriousness unless a criterion is met.
4. Determine expectedness against the product-specific reference safety information, not against general clinical familiarity.
5. A serious case may still be expected.
6. A non-serious case may be unexpected.
7. Serious + unexpected is the key expedited-reporting combination in the supplied training framework.
8. Preserve the evidence and reasoning so another reviewer can reproduce the classification.

10. Final portfolio conclusion

Across the five fictional scenarios, seriousness ranges from routine outpatient findings to life-threatening and disability-level outcomes. Expectedness varies independently because it depends on the product's reference safety information. The exercise therefore demonstrates the central Task 4 principle: **seriousness and expectedness are separate classification dimensions and must be assessed independently before being combined.**

> **Final declaration:** This Task 4 report is a fictional educational training project prepared for a pharmacovigilance internship. It is not a real ICSR, not a patient record and not clinical advice. In actual pharmacovigilance work, current product labeling, validated seriousness criteria and jurisdiction-specific reporting requirements must be used.

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